

应用系统分布式构建运维-数据库 (2026.5.16)

一、环境准备需要 2 个节点，通过 base 克隆出系统出两个系统 mysql1 和 mysql2 保证虚拟网卡的开启且相同模式，我们使用 NAT 模式。

//将 VMnet 的 NAT 网络设置为 192.168.200.0/24 目的是与实验文档保持一致。

修改 IP 地址： mysql1: 192.168.126.10——主节点 mysql2:
192.168.126.20——从节点

手动挂载光驱：

```
mount /dev/cdrom /media
```

也可以让光驱在系统启动时自动挂载：(defaults 0 0 和上面的是一行，mount -a 是在系统中输入的命令)

```
开机自动挂载： vi /etc/fstab
最后一行添加： /dev/cdrom /mnt/cd iso9660
defaults      0 0
mount -a
```

修改主机名：

第一台：mysql1

```
hostnamectl set-hostname mysql1
```

第二台：mysql2

```
hostnamectl set-hostname mysql2
```

修改主机解析：

```
vi /etc/hosts
```

在里面均添加 2 行，写本机 IP 空格+短主机名+长主机名

```
192.168.126.10 mysql1 mysql1.shenzhi.com
```

```
192.168.126.20 mysql2 mysql2.shenzhi.com
```

关闭防火墙：

```
systemctl stop firewalld
```

```
systemctl disable firewalld
```

systemctl status firewalld (检查状态)

```
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)
   Active: inactive (dead)

Nov 28 22:27:50 mysql1 systemd[1]: Stopped firewalld - dynamic firewall daemon.
[root@mysql1 ~]#
```

将 SELINUX 后面的参数修改为“disabled”

```
3. 192.168.220.30 (1) 4. 192.168.220.40 (2) 1. Home 2. 1
# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
SELINUX=disabled
# SELINUXTYPE= can take one of three two values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

setenforce 0 systemctl

关闭 SELinux: (模板制作的时候已经关闭了)

setenforce 0 临时关闭 vi

/etc/selinux/config

SELINUX=enforcing 修改为 disabled

修改错误进入修复模式 (在开机时按
e 进入修复模式)

在 quiet 后面加 selinux=0 (必加)

systemd.unit=rescue.target (可加)

```
load_video
set gfx_payload=keep
insmod gzio
linux ($root)/vmlinuz-4.18.0-553.el8_10.x86_64 root=/dev/mapper/rhel-root ro c\
rashkernel=auto resume=/dev/mapper/rhel-swap rd.lvm.lv=rhel/root rd.lvm.lv=rhe\
l/swap rhgb quiet selinux=0_
initrd ($root)/initramfs-4.18.0-553.el8_10.x86_64.img $tuned_initrd
```

如果写末尾没用需要写中间 (老版本)!!!!!!

二、 安装数据库 mariadb

mariadb 是 mysql 数据库的一个分支。

两个节点均安装 mariadb

yum clean all

yum repolist

yum install mariadb mariadb-server -y

两台均设置为数据库服务自启动

systemctl start mariadb

systemctl enable mariadb

```
[root@mysql1 ~]# systemctl start mariadb
[root@mysql1 ~]# systemctl enable mariadb
Created symlink from /etc/systemd/system/multi-user.target.wants/mariadb.service to /usr/lib/systemd/sy\
stem/mariadb.service.
[root@mysql1 ~]#
```

三、 初始化数据库初始化数据库 mysql_secure_installation 设置密码

为:

关闭远程登录: Disallow root login remotely? [Y/n] n

```
Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] n
... skipping.
```

细节如下:

```
[root@mysql2 ~]# mysql_secure_installation
/usr/bin/mysql_secure_installation: line 379: find_mysql_client: command not found
```

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current password
for the root user. If you've just installed MariaDB, and you haven't set
the root password yet, the password will be blank, so you should just
press enter here.

Enter current password for root (enter for none): OK, successfully
used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB root user
without the proper authorisation.

Set root password? [Y/n] New password:

Re-enter new password:

Password updated successfully!

Reloading privilege tables..

... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation go a bit
smoother. You should remove them before moving into a production
environment.

```
Remove anonymous users? [Y/n] y ...
Success!

Normally, root should only be allowed to connect from 'localhost'. This ensures that
someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] n ...
skipping.

By default, MariaDB comes with a database named 'test' that anyone can access. This
is also intended only for testing, and should be removed before moving into a
production environment.

Remove test database and access to it? [Y/n] y -
Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far will take effect
immediately.

Reload privilege tables now? [Y/n] y ...
Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB installation should
now be secure.

Thanks for using MariaDB!
[root@mysql2 ~]#
```

解析：初始化数据库安全配置，删除匿名用户、测试库，并设置 root 密码。

配置 mysql1 **主节点**（特别注意是哪个节点）修改

mysql1 节点的数据库配置文件。

```
cp /etc/my.cnf.d/server.cnf /etc/my.cnf.d/server.cnf_bak0 vi
/etc/my.cnf.d/server.cnf
```

在配置文件 vi /etc/my.cnf.d/server.cnf 中的[mysqld]增添四行内容：

```
[mysqld]
log_bin = mysql-bin
binlog_ignore_db = mysql
server_id = 10
bind-address = 0.0.0.0
```

- log_bin: 开启二进制日志, 主从复制的基础。
- server_id: 集群中唯一标识, 主节点设为 10 (取自 IP 末段)。
- bind-address = 0.0.0.0: 监听所有网络接口, 允许远程连接。

同一个集群里面, server_id 是不同的。

重启数据库

```
systemctl restart mariadb
```

登录数据库并授权

```
mysql -uroot -p12345678 主节点
```

mysql1 修改: (进入 sql)

在主节点 mysql1 节点, 授权在任何客户端机器上可以以 root 用户登录到数据库
 GRANT REPLICATION SLAVE ON *.* TO 'user'@'%' IDENTIFIED BY '12345678';

重启 **mysql1** 中的 **mariadb** 服务:

```
systemctl restart mariadb
```

分节点 mysql2 修改:

复制 mysql1 中的 vi /etc/my.cnf.d/server.cnf 到 mysql2 中 (切记修改 server_id = 20):

```
[mysqld]
log_bin = mysql-bin
binlog_ignore_db = mysql
server_id = 20
bind-address = 0.0.0.0
解析同上
```

重启数据库

```
systemctl restart mariadb
```

```
mysql -uroot -p12345678
```

在主节点 mysql1 上创建一个 user 用户连接节点 mysql2, 并赋予从节点同步主节点数据库的权限。

```
CHANGE MASTER TO
  MASTER_HOST='192.168.126.10',
  MASTER_USER='user',
  MASTER_PASSWORD='12345678',
  MASTER_PORT=3306;
START SLAVE;
SHOW SLAVE STATUS\G
```

- CHANGE MASTER TO 指定主库地址、复制账号和密码。

- START SLAVE 启动 IO 线程和 SQL 线程。
- SHOW SLAVE STATUS\G 查看状态，确保 Slave_IO_Running 和 Slave_SQL_Running 均为 Yes。

```
MariaDB [(none)]> start slave;
Query OK, 0 rows affected, 1 warning (0.00 sec)

MariaDB [(none)]> show slave status\G
***** 1. row *****
      Slave_IO_State: Waiting for master to send event
      Master_Host: mysql1
      Master_User: user
      Master_Port: 3306
      Connect_Retry: 60
      Master_Log_File: mysql-bin.000002
      Read_Master_Log_Pos: 245
      Relay_Log_File: mariadb-relay-bin.000004
      Relay_Log_Pos: 529
      Relay_Master_Log_File: mysql-bin.000002
      Slave_IO_Running: Yes
      Slave_SQL_Running: Yes
      Replicate_Do_DB:
      Replicate_Ignore_DB:
      Replicate_Do_Table:
      Replicate_Ignore_Table:
      Replicate_Wild_Do_Table:
      Replicate_Wild_Ignore_Table:
      Last_Errno: 0
      Last_Error:
      Skip_Counter: 0
      Exec_Master_Log_Pos: 245
      Relay_Log_Space: 1532
      Until_Condition: None
```

补充：

如果出现 Slave_IO_Running: Connecting 错误

- 检查主节点防火墙、bind-address、网络连通性。
- 查看错误日志：tail -50 /var/log/mariadb/mariadb.log
- 常见原因：主节点未监听公网、密码错误、防火墙拦截。

mysql -uroot -p12345678 -e "SHOW SLAVE STATUS\G" (外部直接看)

测试用例

```
mysql -uuser -p12345678 -h192.168.126.10 P3306 --
protocol=tcp --connect-timeout=10 -e "SELECT 1"
```

重写方式

STOP SLAVE;

RESET SLAVE ALL;

CHANGE MASTER TO

MASTER_HOST='192.168.126.10',

MASTER_USER='user',

```
MASTER_PASSWORD='12345678',
MASTER_PORT=3306;
START SLAVE;
SHOW SLAVE STATUS\G
```

主

1. 确认 MariaDB 监听的地址和端口

```
ss -tlnp | grep 3306
```

2. 确认没有 skip-networking 选项

```
grep -E "^skip-networking|^bind-address" /etc/my.cnf
/etc/my.cnf.d/*.cnf
```

3. 查看 MariaDB 错误日志中关于绑定的信息

```
grep -i "bind\|listen"
/var/log/mariadb/mariadb.log | tail -5
```

4. 临时用 root 从本机连接测试（确保服务正常）

```
mysql -uroot -p12345678 -e "SELECT 1"
```

从（如果这里成功就连接成功，前两种默认没安装）

1. 使用 nc 测试端口是否可达（超时 3 秒）

```
nc -zv -w 3 192.168.126.10 3306
```

2. 使用 telnet（如果没有就安装 yum install telnet -y）

```
telnet 192.168.126.10 3306
```

3. 使用 mysql 客户端强制 TCP 协议并增加超时

```
mysql -uuser -p12345678 -h192.168.126.10 P3306
--protocol=tcp --connect-timeout=10 -e
```

四、验证主从同步建议：验证前先对系统做一个快照。

先在主节点 mysql1 中创建库 test，并在库 test 中创建表 company，插入表数据，创建完成后，查看表 company 数据，命令如下：

```
[root@mysql1 ~]# mysql -uroot -p12345678
MariaDB [(none)]> create database test;
MariaDB [(none)]> use test;
```



```
MariaDB [test]> create table company(id int not null primary key,name varchar(50),addr varchar(255));
MariaDB [test]> insert into company values(1,"alibaba","china");
MariaDB [test]> select * from company;
```

```
[root@mysql1 ~]# mysql -uroot -p000000
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 5.5.44-MariaDB-log MariaDB Server

Copyright (c) 2000, 2015, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database test;
Query OK, 1 row affected (0.00 sec)

MariaDB [(none)]> use test;
Database changed
MariaDB [test]> create table company(id int not null primary key,name varchar(50),addr
-> varchar(255));
Query OK, 0 rows affected (0.00 sec)

MariaDB [test]> insert into company values(1,"alibaba","china");
Query OK, 1 row affected (0.00 sec)

MariaDB [test]>
MariaDB [test]> select * from company;
+-----+-----+-----+
| id | name   | addr  |
+-----+-----+-----+
| 1  | alibaba | china |
+-----+-----+-----+
1 row in set (0.00 sec)

MariaDB [test]> █
```

从节点验证复制功能

登录 mysql2 中的 mysql 数据库 show databases;

可以看到 test 已经自动由主服务器传了过来

```
[root@mysql2 ~]# mysql -uroot -p000000
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 5.5.44-MariaDB-log MariaDB Server

Copyright (c) 2000, 2015, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| test |
+-----+
4 rows in set (0.02 sec)

MariaDB [(none)]> █
```

此篇结束。